

# Jiaxin (Jeremiah) Pan

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## Research Interests

Physics undergraduate working on AI for Science, generative sampling of physical systems, and interpretable crystal descriptors. Interested in statistical mechanics, reproducible scientific computing, and reliable machine-learning methods.

## Education

### 2023–2027 **B.Sc. in Physics (with honours) — Expected**

The Chinese University of Hong Kong, Shenzhen

GPA: 3.61/4.0. Four-year degree with a required research thesis. Admitted to Economics in 2023; completed Year 1 foundation courses before transferring to Physics in 2024.

### 2026 (Spring) **Visiting Student**

University of Auckland · New Zealand

Completed coursework in Physics, Computer Science, and AI, including COMPSCI 361 and COMPSCI 220.

### 2025 (Summer) **Summer Visiting Student**

University of California, Berkeley Extension · USA

Completed coursework in physics and neuroscience.

## Current Research

### Ongoing **Data-Free Generative Sampling of the Two-Dimensional XY Model**

Undergraduate Researcher, CUHK-Shenzhen

Supervised by Prof. Kai Zhou. Provisional portfolio title.

- Contributed to adapting Stochastic Path Sampling (SPS) to periodic XY angles, using energy and forward/backward path probabilities without pre-generated equilibrium training configurations; evaluated paths with path-space independence Metropolis–Hastings (MH) correction.
- Completed a reproducible training-budget experiment across three seeds and three generation-step settings in Python/PyTorch on Slurm-managed GPUs; Wolff sampling provided an independent validation reference.
- Evaluated physical statistics, effective sample size per sampling second, and rejection streaks. Early efficiency gains were consistent; later changes were seed dependent. The limited 4×4, single-temperature study does not establish thermodynamic critical behavior or an advantage over Wolff.

### Ongoing **Interpretable Crystal Representations for AI-Assisted Materials Design**

HSSRLM — AI for Science

Project framework: hierarchical descriptors linking natural language and porous materials, with T5 (Text-to-Text Transfer Transformer) bidirectional translation infrastructure. Broader dataset validation and model evaluation remain in progress.

- Contributed to evidence-aware Crystallographic Information File (CIF)-to-descriptor preprocessing, preserving provenance and separating known, inferred, and missing information.
- Contributed to candidate validation, node/linker/topology connectivity checks, explicit failure reporting, and descriptor-level duplicate tracking.
- Contributed to limited structure-equivalence audits on three publicly sourced metal–organic frameworks, examining information lost by coarse descriptors.

Outputs are descriptors, not complete atomistic CIF structures; rule compatibility does not establish physical validity.

## Additional Research Experience

2026 (Spring) **Remote Research Assistant**

INSAIT · Bulgaria / Remote

Under Dr. Jinjin Gu: technical analysis of large language model reasoning and post-training, including Surgical Post-Training (SPOT), knowledge retention, and regularization; prepared research syntheses and contributed to seminars.

2025 (Fall) **Guided Research on Van der Waals Integration and 2D Electronics**

CUHK-Shenzhen, Prof. Qi Qian's Lab

Hands-on fabrication and characterization of  $\text{MoS}_2$  devices using dry transfer and van der Waals integration; studied metal–semiconductor contacts and polarity modulation. Conducted literature review and prepared an end-of-term report.

2025 (Spring) **Computational Physics Course Project**

CUHK-Shenzhen

Group study of the quantum harmonic oscillator using finite-difference discretization and matrix eigenvalue methods; compared numerical and analytical results, examined convergence, and synthesized the final report in LaTeX.

## Conference Presentation

31 Jan 2026 **CUHK-Shenzhen Undergraduate Research Conference (URC2026)**

Second Undergraduate Research Conference

Presented “Self-Confidence in Math and STEM Major Choice at CUHK-Shenzhen” at my first research conference. Presentation certificate available on the portfolio.

## Teaching

Oct–Dec 2025 **Undergraduate Student Teaching Fellow (USTF)**

CUHK-Shenzhen School of Science and Engineering

Provided tutoring through the Student Help Room, a form of USTF service, for CSC1005 (Introduction to Computer Engineering), CSC1001 (Introduction to Computer Science), and PHY1001 (Mechanics), 23 October–12 December 2025.

## Honours and Certification

2023–2026 Full-Tuition Scholarship, CUHK-Shenzhen; awarded based on the 2023 Gaokao.

2024–2026 Dean's List: School of Management and Economics (AY2023–2024); School of Science and Engineering (AY2024–2025 and AY2025–2026). The 2026 certificate is pending issuance.

2024–2025 Shaw Overcoming Adversity Award, CUHK-Shenzhen Shaw College.

2025 HCIA-AI: Huawei Certified ICT Associate — Artificial Intelligence; valid through December 2028.

## Skills and Languages

Computing Python, PyTorch, NumPy, SciPy, Matplotlib, Git, LaTeX; GPU experiment workflows through Slurm.

Languages English: TOEFL iBT 100 (October 2025); Mandarin: native; Cantonese: basic conversational ability.